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Notice that Hack is not a computing system with tangible hardware. It is a computer system simulated on your existing computer system. A person developing the game Tetris (a popular tile-matching puzzle) on top of the simulated Hack computer system developed from scratch learns a lot about the actual development of a real computer. So, in this project, instead of developing a physical computer from basic hardware, a student develops a simulated computer from basic simulated components to learn about various aspects of computer science.

Now let us discuss a bit more about the working of this project such as the simulated hardware, the machine language, assembly language, the operating system, etc. As we mentioned earlier, the simulated computer is called Hack, a 16-bit machine with its own CPU. The project also deals with a Hardware Description Language to describe the chips of the Hack computer. The Hack computer also has an assembly language of its own. The operating system running on a Hack computer is called Jack. There is a high-level programming language, also called Jack, associated with the Jack OS. The project culminates with the development of application programs on top of the Jack OS, including the popular video game called Tetris. The URL *https://www.nand2tetris.org/copy-of-talks* shows a number of interesting applications developed using Jack, the programming language.

Though we have already mentioned the project 'From Nand to Tetris', in this article we will be focusing on the workings of an educational operating system that is similar to this project. This is an experimental operating system called eXpOS developed by Dr Murali Krishnan K. and his team from the National Institute of Technology, Calicut (NIT-C). eXpOS has many advantages over the Jack OS from the project 'From Nand to Tetris' as far as teaching operating system concepts is concerned. Jack runs on a simulated computer called Hack, which is developed from basic logic gates and has some additional complexity. eXpOS runs on top of a simulated string based architecture called eXperimental String Machine (XSM). A high-level programming language called Experimental Language (EXPL), a system programming language called Systems Programming Language (SPL) and an assembly language called XSM Assembly Language have been used to develop eXpOS and its associated application programs.

eXpOS has a built-in compiler for the high-level programming language EXPL. A detailed manual and course on the development of the EXPL compiler is available at the URL *https://silcnitc.github.io/*. This course was also developed by Dr Murali Krishnan K. and his team from NIT-C. At the URL given earlier, the step-by-step development of the EXPL compiler is given, which is a good learning material for people interested in learning compiler design. However, we will be focusing on the operating system eXpOS, the first of its kind from India.

Before we proceed any further, do note that eXpOS works on top of a simulated hardware called XSM architecture and cannot be installed on real hardware. Similarly, the high-level programming language EXPL, the system programming language SPL and the XSM assembly language only work on eXpOS and XSM architecture, and cannot be used on any real operating system or real hardware. This architecture-neutral behaviour of the XSM machine might look like a weakness to casual observers but, in reality, eXpOS draws a lot of its strength from this simplified XSM architecture. Because of this simplified XSM machine, the development of eXpOS is relatively easy compared with the development of any lightweight operating system. But how can we compare eXpOS with the popular Hack computer. In order to develop a Hack computer, we have to start with simulated NAND gates and proceed forward, whereas eXpOS works on a simulated XSM machine which is supported by experimental programming languages like EXPL, SPL and XSM assembly language. The project 'From Nand to Tetris' covers the entirety of computer science, whereas the project eXpOS teaches just one core field of computer science -- operating systems. Thus, even though the two projects share the same philosophy, they are in no way related and the emphasis of eXpOS on teaching operating systems concepts alone makes it absolutely relevant.

The details of the course on eXpOS are available at the URL *http://exposnitc.github.io/*. There is a detailed roadmap on this URL about the step-by-step development of the operating system eXpOS. It was derived from an earlier experimental operating system called XOS, also developed by the same team from NIT-C. Figure 1 in the previous page is the logo of eXpOS.

eXpOS and all the associated learning material are licensed under the Creative Commons Attribution-NonCommercial 4.0 International License. The development of eXpOS is done in 28 stages, which are divided into three phases. The first phase involves Stages 1 to 12 and helps you to get familiar with the installation, the bootstrap loading process, the XEXE executable file